

SIMVACOR[®] FILM COATED TABLET

Ingredient(s):

Simvacor Film Coated Tablet 20mg:	
Each tablet contains:	
Simvastatin	20mg
Simvacor Film Coated Tablet 40mg:	
Each tablet contains:	
Simvastatin	40mg

Pharmacodynamics:

After oral ingestion, simvastatin is hydrolyzed in the liver to the corresponding active beta-hydroxyacid form which has a potent activity in inhibiting HMG-CoA reductase. This enzyme catalyses the conversion of HMG-CoA to mevalonate, an early and rate-limiting step in the biosynthesis of cholesterol.

Simvastatin reduces both normal and elevated LDL-C concentrations. LDL is formed from very-low-density protein (VLDL) and is catabolised predominantly by the high affinity LDL receptor. Simvastatin may involve both reduction of VLDL-cholesterol (VLDL-C) concentration and induction of the LDL receptor, leading to reduced production and increased catabolism of LDL-C. Apolipoprotein B also falls substantially during treatment with simvastatin. In addition, simvastatin moderately increases HDL-C and reduces plasma TG. As a result of these changes the ratios of total- to HDL-C and LDL- to HDL-C are reduced.

Pharmacokinetics:

Simvastatin is readily hydrolyzed *in vivo* to the corresponding beta-hydroxyacid, a potent inhibitor of HMG-CoA reductase. Hydrolysis takes place mainly in the liver; the rate of hydrolysis in human plasma is very slow.

Absorption

Simvastatin is well absorbed and undergoes extensive hepatic first-pass extraction. The extraction in the liver is dependent on the hepatic blood flow. The liver is the primary site of action of the active form. The availability of the beta-hydroxyacid to the systemic circulation following an oral dose of simvastatin was found to be less than 5 % of the dose. Maximum plasma concentration of active inhibitors is reached approximately 1-2 hours after administration of simvastatin. Concomitant food intake does not affect the absorption. No accumulation of simvastatin occurred after multiple dosing.

Distribution

The protein binding of simvastatin and its active metabolite is > 95 %.

Elimination

Simvastatin is a substrate of CYP3A4. The major metabolites of simvastatin present in human plasma are the beta-hydroxyacid and four additional active metabolites. 13 % of oral dose was excreted in the urine and 60 % in the faeces within 96 hours. Simvastatin is taken up actively into the hepatocytes by the transporter OATP1B1.

Special populations

Pharmacokinetic data in children and adolescents are not available. Carriers of the SLCO1B1 gene c.521T>C allele have lower OATP1B1 activity. Patients with SLCO1B1 polymorphism there is a risk of increased exposure of simvastatin acid, which may lead to an increased risk of rhabdomyolysis.

Indication(s):

REDUCTIONS IN RISK OF CORONARY HEART DISEASE (CHD) MORTALITY AND CARDIOVASCULAR EVENTS: In patients at high risk of coronary events because of existing CHD, history of stroke or other cerebrovascular disease, simvastatin is indicated to: reduce the risk of total mortality by reducing CHD deaths, reduce the risk of non-fatal myocardial infarction and stroke, reduce the need for coronary and non-coronary revascularization procedures.

HYPERLIPIDEMIA: As an adjunct to diet to reduce elevated total-C, LDL-C, Apo B, and TG, and to increase HDL-C in patients with primary hypercholesterolemia, heterozygous familial hypercholesterolemia or combined (mixed) hyperlipidemia when response to diet and other nonpharmacological measures is inadequate. Simvastatin, therefore, lowers the LDL-C/HDL-C and the total-C/HDL-C ratios.

PEDIATRIC PATIENTS WITH HETEROZYGOUS FAMILIAL HYPERCHOLESTEROLEMIA: As an adjunct to diet to reduce total-C, LDL-C, TG, and Apo B levels in adolescent boys and girls who are at least one year post-menarche, 10-17 years of age, with heterozygous familial hypercholesterolemia (HeFH).

GENERAL RECOMMENDATIONS: Prior to initiating therapy with simvastatin, secondary causes of hypercholesterolemia (e.g. poorly controlled diabetes mellitus, hypothyroidism, nephrotic syndrome, dysproteinemias, obstructive liver disease, other drug therapy, alcoholism) should be identified and treated.

Dosage and Administration:

The dosage range for simvastatin is 5-80 mg/day, given as a single dose in the evening. Adjustments of dosage, if required, should be made at intervals of not less than 4 weeks, to a maximum of 80 mg/day given as a single dose in the evening. The 80-mg dose is only recommended in patients at high risk for cardiovascular complications who have not achieved treatment goals on lower doses and when the benefits are expected to outweigh the potential risks. The recommended usual starting dose is 20-40 mg once a day in the evening.

PATIENTS AT HIGH RISK OF CHD OR WITH EXISTING CHD: The usual starting dose is 40 mg/day given as a single dose in the evening in patients at high risk of CHD (with or without hyperlipidemia), i.e., patients with diabetes, history of stroke or other cerebrovascular disease, peripheral vessel disease, or with existing CHD. Drug therapy can be initiated simultaneously with a standard cholesterol-lowering diet and exercise.

PATIENTS WITH HYPERLIPIDEMIA (WHO ARE NOT IN THE RISK CATEGORIES ABOVE): The patient should be placed on a standard cholesterol-lowering diet before receiving simvastatin and should continue on this diet during treatment with simvastatin. The usual starting dose is 20 mg/day given as a single dose a day in the evening. Patients who require a large reduction in LDL-C (more than 45%) may be started at 40mg/day given as single dose in the evening. Patients with mild to moderate hypercholesterolemia can be treated with a starting dose of 10mg of simvastatin. Adjustments of dosage, if required, should be made as specified above.

HOMOZYGOUS FAMILIAL HYPERCHOLESTEROLEMIA: The recommended dosage for patients with homozygous familial hypercholesterolemia is simvastatin 40 mg/day in the evening. The 80-mg dose is only recommended when the benefits are expected to outweigh the potential risks. Simvastatin should be used as an adjunct to other lipid-lowering treatments (e.g., LDL apheresis) in these patients or if such treatments are unavailable. In patients taking lomitapide concomitantly with simvastatin, the dose of simvastatin should not exceed 40 mg/day.

CONCOMITANT THERAPY: Simvastatin is effective alone or in combination with bile acid sequestrants. In patients taking fibrates (other than gemfibrozil or fenofibrate) concomitantly with Simvacor Film Coated Tablet, the dose of Simvacor Film Coated Tablet should not exceed 10mg/day. In patients taking amiodarone, verapamil, or diltiazem concomitantly with Simvacor Film Coated Tablet, the dose of Simvacor Film Coated Tablet should not exceed 20mg/day. In patients taking amlodipine or lipid-lowering dose of niacin (≥ 1g/day) concomitantly with simvastatin, the dose of Simvacor Film Coated Tablet should not exceed 40 mg/day.

DOSAGE IN RENAL INSUFFICIENCY: Because Simvastatin does not undergo significant renal excretion, modification of dosage should not be necessary in patients with moderate renal insufficiency. In patients with severe renal insufficiency (creatinine clearance <30 mL/min), dosages above 10 mg/day should be carefully considered and, if deemed necessary, implemented cautiously.

Mode of Administration:

For oral use.

Contraindications:

1. Hypersensitivity to the active substance and the product.
2. Active liver disease or unexplained persistent elevations of serum transaminases.
3. Pregnancy and nursing women.
4. Concomitant administration of potent CYP3A4 inhibitors (e.g. itraconazole, ketoconazole, posaconazole, voriconazole, HIV protease inhibitors, boceprevir, telaprevir, erythromycin, clarithromycin, telithromycin and nefazodone).
5. Concomitant administration of gemfibrozil, cyclosporine, or danazol.

Precaution(s) / Warning(s):

1. Myopathy/ Rhabdomyolysis
Simvastatin, like other inhibitors of HMG-CoA reductase, occasionally causes myopathy manifested as muscle pain, tenderness or weakness with creatine kinase (CK) above 10X the upper limit of normal (ULN). Myopathy sometimes takes the form of rhabdomyolysis with or without acute renal failure secondary to myoglobinuria, and very rare fatalities have occurred. The risk of myopathy is increased by high levels of HMG-CoA reductase inhibitory activity in plasma.

As with other HMG-CoA reductase inhibitors, the risk of myopathy/rhabdomyolysis is dose related.

Reduced function of transport protein

Reduced function of hepatic OATP transport proteins can increase the systemic exposure of simvastatin acid and increase the risk of myopathy and rhabdomyolysis. Reduced function can occur as the result of inhibition by interacting medicines (e.g. ciclosporin) or in patients who are carriers of the SLCO1B1 c.521T>C genotype.

Creatine Kinase measurement

Creatine Kinase (CK) should not be measured following strenuous exercise or in the presence of any plausible alternative cause of CK increase as this makes value interpretation difficult. If CK levels are significantly elevated at baseline (> 5 x ULN), levels should be re-measured within 5 to 7 days later to confirm the results.

Before the treatment

All patients starting therapy with simvastatin, or whose dose of simvastatin is being increased, should be advised of the risk of myopathy and told to report promptly any unexplained muscle pain, tenderness or weakness.

Caution should be exercised in patients with pre-disposing factors for rhabdomyolysis. In order to establish a reference baseline value, a CK level should be measured before starting a treatment in the following situations:

- Elderly (age $\geq$ 65 years)
- Female gender
- Renal impairment
- Uncontrolled hypothyroidism
- Personal or familial history of hereditary muscular disorders
- Previous history of muscular toxicity with a statin or fibrate
- Alcohol abuse.

In such situations, the risk of treatment should be considered in relation to possible benefit, and clinical monitoring is recommended. If a patient has previously experienced a muscle disorder on a fibrate or a statin, treatment with a different member of the class should only be initiated with caution. If CK levels are significantly elevated at baseline ($> 5 \times$ ULN), treatment should not be started.

Whilst on treatment

If muscle pain, weakness or cramps occur whilst a patient is receiving treatment with a statin, their CK levels should be measured. If these levels are found, in the absence of strenuous exercise, to be significantly elevated ($> 5 \times$ ULN), treatment should be stopped. If muscular symptoms are severe and cause daily discomfort, even if CK levels are $< 5 \times$ ULN, treatment discontinuation may be considered. If myopathy is suspected for any other reason, treatment should be discontinued.

There have been very rare reports of an immune-mediated necrotizing myopathy (IMNM) during or after treatment with some statins. IMNM is clinically characterized by:

- persistent proximal muscle weakness and elevated serum creatine kinase, which persist despite discontinuation of statin treatment;
- muscle biopsy showing necrotizing myopathy without significant inflammation;
- improvement with immunosuppressive agents.

If symptoms resolve and CK levels return to normal, then re-introduction of the statin or introduction of an alternative statin may be considered at the lowest dose and with close monitoring.

Periodic CK measurements are recommended as they may be useful to identify subclinical cases of myopathy. However, there is no assurance that such monitoring will prevent myopathy.

Therapy with simvastatin should be temporarily stopped a few days prior to elective major surgery and when any major medical or surgical condition supervenes.

Measures to reduce the risk of myopathy caused by medicinal product interactions (see also Interaction with Other Medicaments)

The risk of myopathy and rhabdomyolysis is significantly increased by concomitant use of simvastatin with potent inhibitors of CYP3A4 (such as itraconazole, ketoconazole, posaconazole, voriconazole, erythromycin, clarithromycin, telithromycin, HIV protease inhibitors (e.g. nelfinavir), boceprevir, telaprevir, nefazodone, medicinal products containing cobicistat), as well as gemfibrozil, ciclosporin, and danazol. Use of these medicinal products is contraindicated.

The risk of myopathy and rhabdomyolysis is also increased by concomitant use of amiodarone, amlodipine, verapamil, or diltiazem with certain doses of simvastatin. The risk of myopathy, including rhabdomyolysis, may be increased by concomitant administration of fusidic acid with statins. For patients with HoFH, this risk may be increased by concomitant use of lomitapide with simvastatin.

Consequently, regarding CYP3A4 inhibitors, the use of simvastatin concomitantly with itraconazole, ketoconazole, posaconazole, voriconazole, HIV protease inhibitors (e.g. nelfinavir), boceprevir, telaprevir, erythromycin, clarithromycin, telithromycin, nefazodone, and medicinal products containing cobicistat is contraindicated. If treatment with potent CYP3A4 inhibitors (agents that increase AUC approximately 5 fold or greater) is unavoidable, therapy with simvastatin must be suspended (and use of an alternative statin considered) during the course of treatment. Moreover, caution should be exercised when combining simvastatin with certain other less potent CYP3A4 inhibitors: fluconazole, verapamil, diltiazem. Concomitant intake of grapefruit juice and simvastatin should be avoided.

The use of simvastatin with gemfibrozil is contraindicated. Due to the increased risk of myopathy and rhabdomyolysis, the dose of simvastatin should not exceed 10 mg daily in patients taking simvastatin with other fibrates, except fenofibrate. Caution should be used when prescribing fenofibrate with simvastatin, as either agent can cause myopathy when given alone.

Simvastatin must not be co-administered with systemic formulations of fusidic acid or within 7 days of stopping fusidic acid treatment. In patients where the use of systemic fusidic acid is considered essential, statin treatment should be discontinued throughout the duration of fusidic acid treatment. There have been reports of rhabdomyolysis (including some fatalities) in patients receiving fusidic acid and statins in combination. The patient should be advised to seek medical advice immediately if they experience any symptoms of muscle weakness, pain or tenderness. Statin therapy may be re-introduced seven days after the last dose of fusidic acid. In exceptional circumstances, where prolonged systemic fusidic acid is needed, e.g., for the treatment of severe infections, the need for co-administration of simvastatin and fusidic acid should only be considered on a case by case basis and under close medical supervision.

The combined use of simvastatin at doses higher than 20 mg daily with amiodarone, amlodipine, verapamil, or diltiazem should be avoided. In patients with HoFH, the combined use of simvastatin at doses higher than 40 mg daily with lomitapide must be avoided.

Patients taking other medicines labelled as having a moderate inhibitory effect on CYP3A4 concomitantly with simvastatin, particularly higher simvastatin doses, may have an increased risk of myopathy. When coadministering simvastatin with a moderate inhibitor of CYP3A4 (agents that increase AUC approximately 2-5 fold), a dose adjustment of simvastatin may be necessary. For certain moderate CYP3A4 inhibitors e.g. diltiazem, a maximum dose of 20mg simvastatin is recommended.

Simvastatin is a substrate of the Breast Cancer Resistant Protein (BCRP) efflux transporter. Concomitant administration of products that are inhibitors of BCRP (e.g., elbasvir and grazoprevir) may lead to increased plasma concentrations of simvastatin and an increased risk of myopathy; therefore, a dose adjustment of simvastatin should be considered depending on the prescribed dose. Co-administration of elbasvir and grazoprevir with simvastatin has not been studied; however, **the dose of simvastatin should not exceed 20 mg daily in patients receiving concomitant medication with products containing elbasvir or grazoprevir.**

Rare cases of myopathy/rhabdomyolysis have been associated with concomitant administration of HMG-CoA reductase inhibitors and lipid-modifying doses (≥ 1 g/day) of niacin (nicotinic acid), either of which can cause myopathy when given alone.

2. Hepatic effects

Persistent increases (to $> 3 \times$ ULN) in serum transaminases have occurred in a few adult patients who received simvastatin. When the drug was interrupted or discontinued in these patients, the transaminase levels usually fell slowly to pre-treatment levels.

It is recommended that liver function tests be performed before treatment begins and thereafter when clinically indicated. Patients titrated to the 80-mg dose should receive an additional test prior to titration, 3 months after titration to the 80-mg dose, and periodically thereafter (e.g., semi-annually) for the first year of treatment. Special attention should be paid to patients who develop elevated serum transaminase levels, and in these patients, measurements should be repeated promptly and then performed more frequently. If the transaminase levels show evidence of progression, particularly if they rise to $3 \times$ ULN and are persistent, the drug should be discontinued. Note that ALT may emanate from muscle, therefore ALT rising with CK may indicate myopathy.

The product should be used with caution in patients who consume substantial quantities of alcohol.

As with other lipid-lowering agents, moderate ($< 3 \times$ ULN) elevations of serum transaminases have been reported following therapy with simvastatin. These changes appeared soon after initiation of therapy with simvastatin, were often transient, were not accompanied by any symptoms and interruption of treatment was not required.

3. Diabetes mellitus

Statins may raise blood glucose and in some patients, at high risk of future diabetes, may produce a level of hyperglycaemia where formal diabetes care is appropriate. This risk, however, is outweighed by the reduction in vascular risk with statins and therefore should not be a reason for stopping statin treatment. Patients at risk (fasting glucose 5.6 to 6.9mmol/L, BMI $>30\text{kg/m}^2$, raised triglycerides, hypertension) should be monitored both clinically and biochemically according to national guidelines.

4. Interstitial lung disease

Statins may cause interstitial lung disease, especially with long term therapy. Presenting features can include dyspnoea, non-productive cough and deterioration in general health (fatigue, weight loss and fever). If it is suspected a patient has developed interstitial lung disease, statin therapy should be discontinued.

5. Paediatric population

Safety and effectiveness of simvastatin in patients 10-17 years of age with heterozygous familial hypercholesterolemia have been evaluated in adolescent boys and in girls who were at least one year post-menarche. Doses greater than 40 mg have not been studied in this population. There was no detectable effect on growth or sexual maturation in the adolescent boys or girls, or any effect on menstrual cycle length in girls. Adolescent females should be counselled on appropriate contraceptive methods while on simvastatin therapy. Simvastatin has not been studied in patients younger than 10 years of age, nor in pre-menarchal girls.

6. Exciipient

This product contains lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

7. Effects on Ability to Drive and Use Machines

Simvastatin has no or negligible influence on the ability to drive and use machines. However, when driving vehicles or operating machines, it should be taken into account that dizziness has been rarely reported.

8. **Myasthenia Gravis/ Ocular Myasthenia**

In few cases, statins have been reported to induce de novo or aggravate pre-existing myasthenia gravis or ocular myasthenia. Simvacor® Film Coated Tablet should be discontinued in case of aggravation of symptoms. Recurrences when the same or a different statin was (re-) administered have been reported.

Interaction with Other Medicaments:

Prescribing recommendations for interacting agents are summarized in the table below:

Drug Interactions Associated with Increased Risk of Myopathy/Rhabdomyolysis	
Interacting agents	Prescribing recommendations
Potent CYP3A4 inhibitors, e.g.: Itraconazole Ketoconazole Posaconazole Voriconazole Erythromycin Clarithromycin Telithromycin HIV protease inhibitors (e.g. nelfinavir) Boceprevir Telaprevir Nefazodone Cobicistat Ciclosporin Danazol Gemfibrozil	Contraindicated with simvastatin
Other fibrates (except fenofibrate)	Do not exceed 10 mg simvastatin daily
Fusidic acid	Is not recommended with simvastatin
Niacin (nicotinic acid) (≥ 1 g/day)	For Asian patients, not recommended with simvastatin
Amiodarone Amlodipine Verapamil Diltiazem Elbasvir Grazoprevir	Do not exceed 20 mg simvastatin daily
Lomitapide	For patients with HoFH, do not exceed 40 mg simvastatin daily
Grapefruit juice	Avoid grapefruit juice when taking simvastatin

Effects of other medicinal products on simvastatin

Contraindicated drugs

Potent inhibitors of CYP3A4: Concomitant use with medicines labeled as having a potent inhibitory effect on CYP3A4 at therapeutic doses (e.g.: itraconazole, ketoconazole, posaconazole, voriconazole, erythromycin, clarithromycin, telithromycin, HIV protease inhibitors, boceprevir, telaprevir or nefazodone) is contraindicated. If treatment with potent CYP3A4 inhibitors is unavoidable, therapy with simvastatin should be suspended during the course of treatment.

Gemfibrozil, cyclosporine or danazol: Concomitant use of these drugs with simvastatin is contraindicated.

Other Drugs

Other Fibrates

The dose of simvastatin should not exceed 10 mg daily in patients receiving concomitant medication with fibrates other than gemfibrozil or fenofibrate. When simvastatin and fenofibrate are given concomitantly, there is no evidence that the risk of myopathy exceeds the sum of the individual risks of each agent. Caution should be used when prescribing fenofibrate with simvastatin, as either agent can cause myopathy when given alone. Addition of fibrates to simvastatin typically provides little additional reduction in LDL-C, but further reductions of TG and further increases in HDL-C may be obtained. Combinations of fibrates with simvastatin have been used without myopathy in small short-term clinical studies with careful monitoring.

Concurrent use of fibrates may cause severe myositis and myoglobinuria.

Fluconazole

Rare cases of rhabdomyolysis associated with concomitant administration of simvastatin and fluconazole have been reported.

Fusidic acid

The risk of myopathy including rhabdomyolysis may be increased by the concomitant administration of systemic fusidic acid with statins. The mechanism of this interaction (whether it is pharmacodynamics or pharmacokinetic, or both) is yet unknown. There have been reports of rhabdomyolysis (including some fatalities) in patients receiving this combination. Co-administration of this combination may cause increased plasma concentrations of both agents. If treatment with fusidic acid is necessary, simvastatin treatment should be discontinued throughout the duration of the fusidic acid treatment.

Amiodarone

The risk of myopathy and rhabdomyolysis is increased by concomitant administration of amiodarone with simvastatin. In a clinical trial, myopathy was reported in 6% of patients receiving simvastatin 80 mg and amiodarone. The dose of simvastatin should not exceed 20 mg daily in patients receiving concomitant medication with amiodarone.

Calcium Channel Blockers

• **Verapamil**

The risk of myopathy and rhabdomyolysis is increased by concomitant administration of verapamil with simvastatin 40 mg or 80 mg. In a clinical trial, patients on verapamil treated concomitantly with simvastatin 80 mg had an increased risk of myopathy. The dose of simvastatin should not exceed 20 mg daily in patients receiving concomitant medication with verapamil.

• **Diltiazem**

The risk of myopathy and rhabdomyolysis is increased by concomitant administration of diltiazem with simvastatin 80 mg. In a clinical trial, patients on diltiazem treated concomitantly with simvastatin 80 mg had an increased risk of myopathy. The dose of simvastatin should not exceed 20 mg daily in patients receiving concomitant medication with diltiazem.

• **Amlodipine**

Patients on amlodipine treated concomitantly with simvastatin have an increased risk of myopathy. In a clinical trial, patients on amlodipine treated concomitantly with simvastatin 80 mg had a slightly increased risk of myopathy. The dose of simvastatin should not exceed 40 mg daily in patients receiving concomitant medication with amlodipine.

Lomitapide

The risk of myopathy and rhabdomyolysis may be increased by concomitant administration of lomitapide with simvastatin. Therefore, in patients with HoFH, the dose of simvastatin must not exceed 40 mg daily in patients receiving concomitant medication with lomitapide.

Moderate Inhibitors of CYP3A4

Patients taking other medicines labelled as having a moderate inhibitory effect on CYP3A4 concomitantly with simvastatin, particularly higher simvastatin doses, may have an increased risk of myopathy.

Inhibitors of the Transport Protein OATP1B1

Simvastatin acid is a substrate of the transport protein OATP1B1. Concomitant administration of medicinal products that are inhibitors of the transport protein OATP1B1 may lead to increased plasma concentrations of simvastatin acid and an increased risk of myopathy.

Inhibitors of Breast Cancer Resistant Protein (BCRP)

Concomitant administration of medicinal products that are inhibitors of BCRP, including products containing elbasvir or grazoprevir, may lead to increased plasma concentrations of simvastatin and an increased risk of myopathy.

Niacin ≥1g/day

The dose of simvastatin should not exceed 40mg daily in patients receiving concomitant medication with niacin (nicotinic acid) ≥1 g/day. Cases of myopathy/rhabdomyolysis have been observed with simvastatin co-administered with lipid-modifying doses (≥1 g/day) of niacin.

Grapefruit juice

Grapefruit juice inhibits cytochrome P450 3A4. Concomitant intake of large quantities (over 1 litre daily) of grapefruit juice and simvastatin resulted in a 7-fold increase in exposure to simvastatin acid. Intake of 240 ml of grapefruit juice in the morning and simvastatin in the evening also resulted in a 1.9-fold increase. Intake of grapefruit juice during treatment with simvastatin should therefore be avoided.

Colchicine

There have been reports of myopathy and rhabdomyolysis with the concomitant administration of colchicine and simvastatin in patients with renal impairment. Close clinical monitoring of such patients taking this combination is advised.

Rifampicin

Because rifampicin is a potent CYP3A4 inducer, patients undertaking long-term rifampicin therapy (e.g. treatment of tuberculosis) may experience loss of efficacy of simvastatin.

Effects of simvastatin on the pharmacokinetics of other medicinal products

Simvastatin does not have an inhibitory effect on cytochrome P450 3A4. Therefore, simvastatin is not expected to affect plasma concentrations of substances metabolised via cytochrome P450 3A4.

Oral anticoagulants

In patients taking coumarin anticoagulants, prothrombin time should be determined before starting simvastatin and frequently enough during early therapy to ensure that no significant alteration of prothrombin time occurs. Once a stable prothrombin time has been documented, prothrombin times can be monitored at the intervals usually recommended for patients on coumarin anticoagulants. If the dose of simvastatin is changed or discontinued, the same procedure should be repeated. Simvastatin therapy has not been associated with bleeding or with changes in prothrombin time in patients not taking anticoagulants.

Pregnancy and Lactation:

Simvastatin is contraindicated during pregnancy as safety in pregnant women has not been established. Maternal treatment with simvastatin may reduce the foetal levels of mevalonate which is a precursor of cholesterol biosynthesis. Atherosclerosis is a chronic process, and ordinarily discontinuation of lipid-lowering medicinal products during pregnancy should have little impact on the long-term risk associated with primary hypercholesterolaemia. For these reasons, simvastatin must not be used in women who are pregnant, trying to become pregnant or suspect they are pregnant. Treatment with simvastatin must be suspended for the duration of pregnancy or until it has been determined that the woman is not pregnant.

It is not known whether simvastatin or its metabolites are excreted in human milk. Because of the potential for serious adverse reactions, women taking simvastatin should not breast-feed their infants.

Side Effect(s) / Adverse Reaction(s):

Blood and lymphatic system disorders:

Rare: anaemia

Psychiatric disorders:

Very rare: insomnia

Frequency Not Known: depression

Nervous system disorders:

Rare: headache, paresthesia, dizziness, peripheral neuropathy

Very rare: memory impairment

Frequency Not Known: myasthenia gravis

Respiratory, thoracic and mediastinal disorders:

Frequency Not Known: interstitial lung disease

Gastrointestinal disorders:

Rare: constipation, abdominal pain, flatulence, dyspepsia, diarrhoea, nausea, vomiting, pancreatitis

Hepatobiliary disorders:

Rare: hepatitis/jaundice

Very rare: fatal and non-fatal hepatic failure

Skin and subcutaneous tissue disorders:

Rare: rash, pruritus, alopecia

Musculoskeletal and connective tissue disorders:

Rare: myopathy (including myositis), rhabdomyolysis with or without acute renal failure, myalgia, muscle cramps

Frequency Not Known: tendinopathy, sometimes complicated by rupture; immune-mediated necrotizing myopathy (IMNM)

Reproductive system and breast disorders:

Not known: erectile dysfunction

General disorders and administration site conditions:

Rare: asthenia

An apparent hypersensitivity syndrome has been reported rarely which has included some of the following features: angioedema, lupus-like syndrome, polymyalgia rheumatica, dermatomyositis, vasculitis, thrombocytopenia, eosinophilia, ESR increased, arthritis and arthralgia, urticaria, photosensitivity, fever, flushing, dyspnoea and malaise.

Investigations:

Rare: increases in serum transaminases (alanine aminotransferase, aspartate aminotransferase, γ-glutamyl transpeptidase), elevated alkaline phosphatase; increase in serum CK levels.

Increases in HbA1c and fasting blood glucose levels have been reported with statins. The risk of hyperglycemia, however, is outweighed by the reduction in vascular risk with statins.

There have been rare post-marketing reports of cognitive impairment (e.g., memory loss, forgetfulness, amnesia, memory impairment, confusion) associated with statin use. These cognitive issues have been reported for all statins. The reports are generally non-serious, and reversible upon statin discontinuation, with variable times to symptom onset (1 day to years) and symptom resolution (median of 3 weeks).

Eye disorders:

Frequency Not Known: ocular myasthenia

The following additional adverse events have been reported with some statins:

- Sleep disturbances, including nightmares
- Sexual dysfunction
- Diabetes mellitus: Frequency will depend on the presence or absence of risk factors (fasting blood glucose ≥ 5.6 mmol/L, BMI > 30 kg/m², raised triglycerides, history of hypertension).

Paediatric population (ages 10-17 years)

The long-term effects on physical, intellectual, and sexual maturation are unknown. No sufficient data are currently available.

Symptoms and Treatment for Overdosage, and Antidote(s):

A few cases of overdosage have been reported; all patients recovered without sequelae. The maximum dose taken was 3.6g. General measures should be adopted.

Shelf-Life:

2 years from the date of manufacture.

Storage Condition(s):

Store at temperature below 30°C. Protect from light and moisture.

Product Description and Packing(s):

Simvacor Film Coated Tablet 20mg

An orange brown elliptical film coated tablet, one side impressed with score and the other side with a mark "

Blister packing of 10's x 3, 10's x 10, 10's x 12 and 10's x 50.



Simvacor Film Coated Tablet 40mg

A pale orange-red elliptical film coated tablet, one side impressed with score and the other side with a mark "

Blister packing of 10's x 3, 10's x 10, 10's x 12 and 10's x 50.



Manufacturer and Product Registration Holder:
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